

device, heating can be performed by tiny heaters embedded within each memory cell.

The origin

Stanford R. Ovshinsky was the first person to consider using resistive changes in phase-change materials as the basis for storing digital data. He patented this idea in the mid sixties and published a paper on it in 1968, when he owned a company, named Energy Conversion Devices Inc. Ovshinsky later founded another company called Ovonyx to commercialise the technology for memory devices.

For many years the development failed to get much traction because of the unavailability of sufficient funds. During the 1990s, a joint venture between Intel and STMicroelectronics acquired the technology rights from Ovonyx and formed a company called Numonyx in 2008, to further develop the PCM technology.

Numonyx quickly brought out a prototype 256Mb memory device fabricated with 90nm technology. Soon after, Samsung demonstrated a 512Mb PCM random-access memory (PRAM) chip made with an 58nm process technology. Numonyx had difficulty surpassing this offer from its South Korean rival so it subsequently focused on stacking several 128Mb chips to produce higher-capacity devices, although none was commercialised. By 2010, Numonyx was sold to the pure memory manufacturer Micron Technologies of Boise, Idaho, and now Micron and Samsung are the main contenders in the PCM race.

Competing with other technologies

As both the crystalline and amorphous phases are stable, PCMs are non-volatile, i.e., the memory retains its contents even if the power is switched off. It is very interesting to compare the attributes of PCM with Flash memory – the most widespread non-volatile memory technology at this time. Fig. 1 lists pertinent features of the two technologies (also included are dynamic

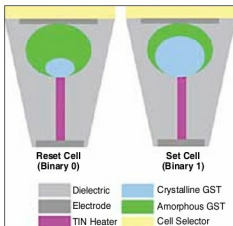


Fig. 2: Structure of a PCM cell in set and reset states

RAM attributes, for reference).

The main drawback of the contemporary Flash memories is their slow access time, which relegates them to be used only as storage devices. They also require erasing of a whole block before even one bit can be changed – a process that needs several milliseconds.

PCMs, on the other hand, feature write speed of around 100 ns and read speed of around 20 ns, which make them blazingly fast. This is all due to the exceptionally fast speed with which the crystalline-amorphous phase change can be affected in materials such as GST.

Flash memories can have very high densities because each memory cell consists of a floating-gate memory transistor tightly coupled with a cell selector transistor, with their combination taking up very little silicon area. The actual phase-change memory cells in a PCM device are very small but the cell-selection transistor required to select each memory cell (bit location) is relatively large. This somewhat limits the achievable integration density, but rapid progress is being made to overcome this limitation and to produce high bit-count PCM chips.

The maximum number of read-write cycles possible with Flash memory is quite limited. Most commercial-grade Flash memory chips are guaranteed for at most 5000 memory cycles because of the wear and tear on memory transistors due to the high programming voltages used.

PCM wins outright here because

the crystalline-amorphous phase changes cause no material degradation. PCM chips are guaranteed for at least one million read-write cycles, which makes them suitable for long-life embedded applications with frequent data changes. Such applications can also benefit from the high data retention time of PCM devices, which exceeds 10 years at ambient temperatures.

Finally, Flash memories are going to run into a fundamental scaling limitation once their minimum feature size gets below 22 nm. It will be hard to scale them down further because memory transistors of such small dimensions hold only a few hundred electrons to signal the state as '0' or '1.' Random thermal fluctuations can flip the logic state, making the memory increasingly unreliable.

PCM devices, in comparison, can be scaled down to five nanometres in minimum feature dimension without any adverse effects because their structure is much simpler and the phases are immune to random thermal fluctuations. This feature has been much discussed in the technical circles and there is some agreement that PCM is the successor to Flash technology – gradually overtaking the latter as the minimum process technology dimension shrinks below 22 nm.

Yet another advantage of PCM stems from the fact that data is stored as a stable structural phase rather than as the presence or absence of charge carriers, as is the case with other current non-volatile memory technologies. This means that PCM is not affected by ionising radiation and is thus useful for military and aerospace applications. This prospect has made defence companies like BAE Systems and Lockheed Martin take serious interest in this memory technology.

Structure and function

Tiny memory cells made from very thin GST films and a titanium-nitride (TiN) heater are fabricated as a memory array on top of complementary metal-oxide semiconductor (CMOS) circuitry for selecting, reading and